

From Molasses to Multi-Feed

Converting seasonal sugar-mill distilleries into 330-day multi-feedstock ethanol plants — the highest-IRR brownfield play in Indian bioenergy

The single-feed trap

Why molasses-only distilleries are the casualties of the consolidation phase

Seasonal by design

A molasses distillery runs on the cane crush: ~180 days of feedstock. Half the year the fermenters sit idle while debt service runs 365 days.

Overcapacity punishes the inflexible

OMCs absorb only ~60% of offered ethanol (CareEdge). Utilisation stuck at 65–75% for 3 years hits single-feedstock plants first — they cannot chase the slab or the season.

Concentration risk everywhere

One feedstock = one price (cane politics set molasses), one policy lever (sugar-diversion caps), one failure mode. The Jul-2023 FCI rice suspension showed how fast a feedstock leg can vanish.

- India's 229 functional cooperative sugar mills are the extreme case: molasses-bound, working-capital-starved (NCDC's ₹10,005 cr scheme is 96.5% working capital), and structurally unable to fund conversion themselves

The opportunity: the slab ladder pays for flexibility

Administered ESY 2024-25 procurement prices — grain campaigns price ₹11-14/L above molasses

Feedstock slab	OMC price (₹/L)	Season
C-heavy molasses	57.97	cane crush only
FCI surplus rice	58.50	allocation-dependent
B-heavy molasses	60.73	cane crush only
Damaged food grains	64.00	year-round
Sugarcane juice/syrup	65.61	cane crush only
Maize	71.86	year-round — the E30 marginal slab

+770

of the 1,212 DFPD-approved projects came in the grain + dual-feed windows (2021-22) — the register already shows the pivot

- The E20→E27/E30 walk adds 400-600 cr L of fuel-ethanol demand (supply-match analysis) — and its marginal litre is grain: conversions capture the increment without greenfield risk
- Multi-feed = slab arbitrage: crush molasses in season, run maize/damaged grain off-season, switch legs as administered prices and allocations move

The policy moat: conversion is the scheme's intended outcome

Every instrument points the same way — flexible, year-round, grain-capable capacity

DFPD Interest Subvention

6% subvention on project loans (grain/dual windows) — debt at ~4.75% effective.
₹1,17,362 cr of loans already recommended across the register.

Administered slab ladder

DA&FW/OMC prices fixed per feedstock per ESY — the maize premium is policy, deliberately steering capacity toward grain.

FCI grain leg

72 LMT rice allocation ESY25-26 at ₹2,320/qtl + damaged-grain channel — a second grain feedstock when maize runs hot.

Long-term offtake

OMC LTOAs + 5% GST + excise-free ethanol molecule; allocation cycles favour plants that can bid multiple slabs.

Cooperative rescue money

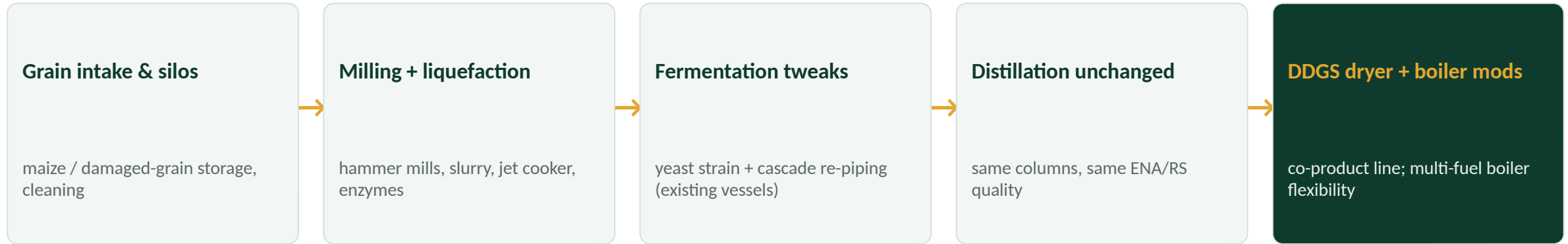
NCDC ₹10,005 cr keeps CSMs alive (working capital) — but only ₹251 cr reached ethanol assets: the conversion capital gap IS the entry.

State layering

UP/Bihar/MP bioenergy policies add capital subsidy and single-window clearances on top (UP added ~50 cr L capacity in 2025 alone).

What actually gets built: ₹50 cr on a live plant

100 KLPD molasses distillery → dual-feed · fermentation and distillation columns stay · ~12-month execution, season-synchronised



Operating days: 180 → 330

- +150 grain days × 100 KLPD = +1.5 cr L/yr incremental ethanol
- Same licence, same LTOA counterparty, same tank farm and ETP headroom

Per-litre grain economics (₹/L)

- Maize slab 71.86 – maize 57.50 (₹23,000/t ÷ 400 L/t) + DDGS credit 13.18 – conversion cost 13.00
- = ₹14.5/L EBITDA — vs ₹8–11/L typical on molasses legs

Financials: brownfield returns greenfield cannot touch

₹50 cr conversion • 70:30 debt:equity • DFPD subvention → 3.5% effective debt • 15-yr life • incremental cash flows only



Sensitivity — project IRR (%): maize price × grain-campaign days

Maize ₹/t	100 days	150 days (base)	200 days
20,000	38.8	56.5	73.2
23,000 (base)	25.8	38.4	50.2
26,000	10.3	18.1	24.9

Honest stress: if grain litres price at the damaged-grain slab (₹64) instead of maize, IRR is still 17.0% — above the 12% hurdle

- The slab premium is ADMINISTERED, not market — the biggest lever and the biggest risk; the ₹26,000-maize row and the ₹64-slab stress are the honest downside reading

Risks and mitigants

The model's own caveats, answered structurally

Risk	Reality	Mitigant
Slab revision	Maize premium is a policy choice, revised every ESY	Multi-feed = switch legs; DFG-slab stress still clears hurdle (17%); slabs have only risen since 2018
OMC allocation	Only ~60% of offered ethanol absorbed today	E27/E30 walk absorbs the glut (utilisation 76–83%); flexible plants win allocation cycles
Maize price	₹26,000/t compresses IRR to 10–25%	DFG + FCI rice as alternate grain legs; forward contracts with FPOs; DDGS hedge (feed prices co-move)
Execution on a live plant	Conversion during crush season kills a year	12-month build synchronised to off-season; proven EPC (Praj-class); fixed-price contract
DDGS market	Credit is ₹13/L of the margin	Poultry-belt siting; take-or-pay feed offtakes; model works at DDGS –30% (IRR ~29%)

- Not relying on: E30 arriving on schedule, carbon revenue, or any slab premium widening — base case is today's published prices

Landscape: the listed players are already converting

The playbook is proven — the gap is the cooperative and mid-tier private mills

Proof it works

Bajaj Hindusthan: 570 KLD new multi-feed (group >1,200 KLPD). Dalmia Bharat Sugar: ₹400 cr grain distillery + two 100-KL units. Balrampur, Triveni, Dhampur all running dual-feed. UP added ~50 cr L in 2025; Gorakhpur GIDA is the grain-ethanol cluster.

EPC capacity exists

Praj Industries anchors multi-feed conversion EPC with standard packages; 12-month brownfield timelines are routine. The DFPD register (1,212 projects) means DPRs, vendors and lenders all know the template.

The gap = the deal flow

229 cooperative mills + dozens of mid-tier private molasses plants have the site, licence, ETP and LTOA — but not the ₹50 cr or the grain-operations capability. NCDC keeps them solvent, not strategic.

- Synergy with the CBG platform pitch: the same mills' press-mud feeds CBG digesters — one partnership, two energy assets, shared site services

The ask: a 5-mill conversion programme

₹250 cr

5 × 100-KLPD conversions (₹75 cr equity)

+7.5 cr L

incremental grain ethanol/yr

₹109 cr

incremental EBITDA/yr at base

Year 3

programme payback

- Structure per mill: BOT/lease or JV on the existing distillery — mill contributes plant, licence and molasses leg; programme funds and operates the grain conversion; revenue share aligned to slabs actually realised
- Phase 1: 2 UP mills (grain belt, GIDA vendor ecosystem, state policy top-up) + 1 Maharashtra cooperative; Phase 2: replicate ×2 with the shared grain-procurement and DDGS-marketing organisation
- Downside protection: DFPD-subsidised debt at ~3.5% effective, DFG/FCI-rice alternate slabs, and the molasses leg keeps servicing debt even if the grain campaign pauses
- Exit: sale to a listed sugar-ethanol consolidator, or the mills buy back the conversion at a pre-agreed multiple once cash flows season